



Granular sludge is a preferable inoculum for the biochemical methane potential assay for two complex substrates

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ABSTRACT

The biochemical methane potential (BMP) assay is a standard method for characterizing biomethane potential and anaerobic biodegradability organic waste streams. Therefore, the BMP protocol must be standardized to reliably compare these parameters for various substrates. Here, the effect of inoculum selection on biomethane potential was investigated through BMP tests using two different substrates and inocula obtained from four different anaerobic digesters. It was found that inocula in the form of granular sludge yielded overall higher biomethane potential and generally had faster kinetics than suspended biomass. Furthermore, acclimation of inocula to substrate appeared to have little effect on degradation rates, and co-inoculation (with both suspended and granular biomass) did not perform better than single inoculation (e.g., with suspended sludge alone). These results emphasize the role of granular sludge as an preferable inoculum for BMP assay.

1. Introduction

Anaerobic digestion (AD) is a well-established technology for bioenergy production with growing applications in agriculture, municipal wastewater treatment, and food processing (Angenent et al., 2018). Because a wide variety of complex, particulate organic wastes are used as substrates for AD — namely animal manure, sewage sludge, and food waste — a standard method for characterizing biomethane potential of substrates is required. The biochemical methane potential (BMP) assay is the preferred method for characterizing the biomethane potential and biodegradable fraction of substrates. Since its initial introduction by Owen et al. (1979), many modifications have been made to the BMP assay in an attempt to standardize and more precisely implement it (Labatut et al., 2011). However, a high degree of variability still exist — especially for complex substrates. For example, Kafle and Chen (2016) found high variability in biomethane potentials and biodegradable fractions for different livestock manures (dairy, chicken, swine, horse, and goat manure), which they correlated the observed variation to the nutrient (N-P-K) profile of each substrate.

Many factors contribute to these variations in final biomethane potential obtained from the BMP assay, namely initial substrate

concentration, particle size, nutrient availability, inoculum-to-substrate ratio, and inoculum selection (Holliger et al., 2016; Koch et al., 2019; Okoro-Shekwaga et al., 2019). The effect of inoculum selection is surprising; as traditional ecological theory suggests that substrate availability alone should dictate final biomethane yield. However, studies have suggested that inoculum selection influences biomethane potential, due to relative abundances of methanogens in the inoculum (De Vrieze et al., 2015). Moreover, production rates were found to be significantly influenced by the inoculum, and it was suggested that selecting an inoculum that is adapted to a particular substrate may improve the kinetics of the process (Koch et al., 2017). Furthermore, inoculum selection appears to influence biomethane production both in the start-up stages of the AD process and during longer retention times (De Vrieze et al., 2015).

This study aims to evaluate the effect of inoculum selection on the BMP assay. Particularly, we examined potential differences in biomethane potential occurring from differences in the characteristics of the selected inoculum, such as those found in granular AD-sludge vs. suspended AD-sludge. Another specific objective of this study was to examine the effect of substrate acclimation on the biodegradation kinetics and final biomethane yield. It is important to better understand

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the role of inoculum selection as a determinant of BMP outcomes. By assessing the effect of different inocula on the BMP for different substrates, the outcomes of the assay could be better characterized, which may lead to an improved, more robust assay.

2. Materials and methods

2.1. Substrates and inocula

Two complex, particulate substrates of distinct physicochemical characteristics were evaluated via the BMP assay: dairy manure (referred to here as manure) and waste activated sludge (referred to here as sludge). Manure was taken from Sunnyside Farm (Scipio Center, NY, USA). Sludge was collected from the secondary clarifier at the Ithaca Area Wastewater Treatment Facility (Ithaca, NY). The inocula were taken from four different AD systems. Effluent from the digesters treating manure and sludge, served as suspended inocula, and were named as follows: (1) manure AD-sludge; and (2) waste activated AD-sludge. Additionally, mixed liquor obtained from two different up-flow anaerobic sludge blanket reactors served as granular inocula, and were named as follows: (3) brewery waste AD-granules (Anheuser-Busch Inbev plant, Baldwinsville, NY); and (4) paper waste AD-granules (Westrock, Solvay Mill, Syracuse, NY). Characterization data of all substrates and inocula is given in [Supplementary data](#). Both substrates were similar in terms of their elementary carbon (31–36%) and nitrogen (~4%). Yet, the chemical oxygen demand (COD) concentration for manure was more than five times higher than for waste activated sludge, mainly because manure consists of a relatively high concentration of lignocellulosic fibers and plant cell wall proteins while waste activated sludge mainly consists of a relatively dilute concentration of microbial biomass proteins.

2.2. Biochemical methane potential (BMP)

The BMP assay was used to determine whether the biomethane yields, biodegradable fractions, and biomethane production rates for each substrate were dependent on the inoculum. The BMP assay used in this study was originally developed by [Owen et al. \(1979\)](#) and later adapted by [Labatut et al. \(2011\)](#). Both manure and sludge were evaluated through a BMP assay with each of the four inocula, yielding eight different experimental conditions. Furthermore, additional BMP assays were performed with a mixture of inocula to test the effect of co-inoculation. Each suspended inoculum (manure AD-sludge and waste activated AD-sludge) were mixed with each granular inoculum (brewery waste AD-granules or paper waste AD-granules) on a 1:1 ratio based on the COD, which yielded four distinct co-inocula.

For the BMP assay, the initial concentration of substrate established in each bottle was 1.5 g COD L⁻¹. All BMP assays were conducted in 250-mL media bottles, where the total sample volume was 200 mL with a headspace volume of 50 mL. A 1:1 inoculum-to-substrate ratio on a COD basis was applied. The inocula were incubated at mesophilic conditions (37 ± 2 °C), for about 24 h before being introduced to each substrate to reduce residual organic matter and thus background methane production. The nutrient trace metal solution was adapted from [Owen et al. \(1979\)](#). Deionized water was flushed with 100% N₂ gas and subsequently added to reach the target sample volume in each bottle. The headspace was flushed with N₂ gas, and the bottle was sealed using a round rubber stopper with a three-way Luer connection stopcock (Cole-Palmer Co., Vernon Hills, IL). Each BMP test was run in triplicate, and underwent an incubation at mesophilic conditions (37 ± 2 °C) for 30 d. The biomethane yield and production rate were normalized to the initial COD added to each bottle (i.e., mL CH₄ g COD⁻¹) and corrected to standard temperature and pressure. For each inoculum, three additional bottles with only inoculum were incubated and their average biomethane production was subtracted to account for the residual organic matter and endogenous biomethane production from the inoculum.

2.3. Analytical methods

The total and soluble COD of each substrate and inocula were analyzed using the closed reflux, titrimetric method according to standard methods ([APHA, 2005](#)). Standard methods were also used to measure total solids (TS) and volatile solids (VS) concentration ([APHA, 2005](#)). Total ammonia nitrogen (TAN) and pH were measured using electrodes (Thermo Fisher Scientific, Waltham, MA). Volumetric gas production during the BMP assay was measured using a borosilicate glass syringe (Chemglass Life Sciences Co., Vineland, NJ), and methane concentrations were determined using a gas chromatograph (SRI 8610C; SRI Instruments, Torrance, CA), which was equipped with a thermal conductivity detector.

2.4. Biomethane yields, biodegradable fraction, and biomethane production rate

The biodegradable fraction (f_D) was calculated from the ratio between the observed biomethane potential (at STP) on a COD basis, determined directly from the BMP assay and the theoretical biomethane potential of 350 mL g COD⁻¹ destroyed ([McCarty, 1964](#)). After obtaining the cumulative biomethane production curves throughout time from the BMP assays, the modified Gompertz equation was used ([Syaichurrozi et al., 2013](#)) to model the biomethane production kinetics:

$$y(t) = B_0 \times \exp \left\{ -\exp \left[\frac{\mu e}{B_0} (t_{50} - t) + 1 \right] \right\} \quad (1)$$

where $y(t)$ is cumulative biomethane production (mL CH₄ g COD⁻¹), B is the biomethane potential (mL CH₄ g COD⁻¹), μ is the maximum production rate of biomethane (mL CH₄ COD⁻¹ d⁻¹), t_{50} is the time to produce 50% of the maximum biomethane (d), e is the mathematical constant (2.7182), and t is the cumulative time for biomethane production (d).

In addition, to identify the actual lag-time or the critical time (t_c) to start producing biomethane, a 2nd derivative of $y(t)$ was calculated and plotted throughout time. The maximum value of the 2nd derivative (the highest difference in production rate) was selected to get the critical time. A detailed description of this methodology is given in [Supplementary data](#).

2.5. Statistical analyses

Statistical analyses were performed with SigmaPlot 12.5 (Systat Software Inc., San Jose, CA, USA). Differences in biomethane yields, biodegradable fractions, and biomethane production rates between treatment conditions were compared by a one-way ANOVA ($p < 0.05$), followed by a *post-hoc* test when differences were significant.

3. Results and discussion

3.1. Inoculum to substrate affinity does not necessarily affect the BMP outcomes

For manure, the inoculum taken from a matching AD system (i.e., one continuously treating dairy manure) did result in greater degradation compared to inoculum that was taken from a non-matching AD system (i.e., one continuously treating waste activated sludge). The biodegradable fraction (f_D) was 34% higher ($p < 0.05$) for manure AD-sludge compared to waste activated AD-sludge ([Fig. 1a](#)), resulting in a biomethane yield of ~140 mL CH₄ per g of COD added, similar to values reported by [Labatut et al., \(2011\)](#) and by [Kalfe and Chen \(2016\)](#). Moreover, with respect to maximum production rate, manure degradation was 17% faster when inoculated with manure AD-sludge than when inoculated with waste activated AD-sludge (coefficient table is given in [Supplementary data](#)). Although the BMP assays using waste activated AD-sludge as an inoculum had a shorter t_{50} value (6.6 d)

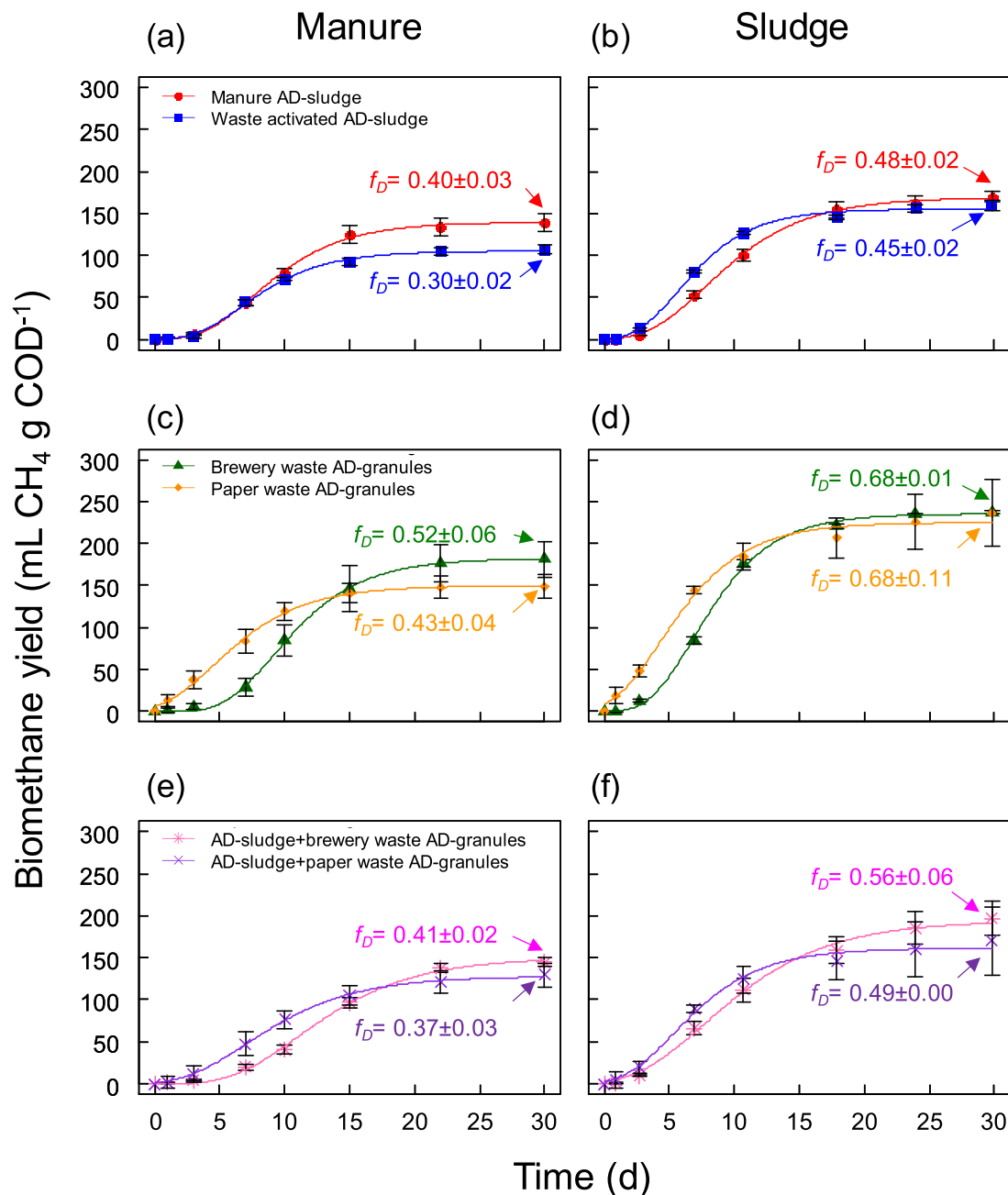


Fig. 1. Time-series curves of dairy manure (left side panels) and waste activated sludge (right side panels) resulting from 30-d BMP assays with different inocula; symbols: observed data; lines: modified Gompertz model fit to data; error bars: standard deviation. For all the regressions, the significance (p) values were lower than 0.0001. Text inside the panels refer to the overall biodegraded fraction (f_D) for each inoculum.

compared to manure AD-sludge (7.8 d), the lag-time, according to the second derivative value (t_c), was similar (2.7–3.3 d). It should be noted that higher final biomethane yields do not necessarily involve faster production rates. For example, Díaz et al. (2016) suggested that for cow manure, even when a higher yield of biomethane is achieved, the production rate might be slower than in cases of lower final biomethane yields.

For sludge, the biodegradable fraction was not significantly different between suspended inocula (Fig. 1b). However, the production rate was 26% lower when the sludge was inoculated with waste activated AD-sludge compared to manure AD-sludge (coefficient table is given in Supplementary data). This was in spite of longer t_{50} and t_c values when manure AD-sludge was used as inoculum (7.7 and 3.6 d, respectively), compared to waste activated AD-sludge (5.6 and 2.3 d, respectively). A similar pattern was observed by Koch

et al. (2017), showing that biomethane yields from sludge were comparable between three different inocula. However, while Koch et al. (2017) suggested that the degradation rate is faster for an inoculum that was already acclimated to the substrate, our data with sludge as substrate and manure AD-sludge as inoculum, shows the opposite. Although both substrates contain different microbial populations, we suspect that any effect imposed by the different biota of the substrates should be similar between inoculum scenarios. Furthermore, the microbial load contained in the substrates was relatively small compared to the inoculum. Since we used a substrate to inoculum ratio of 1:1 (on a COD basis), proportionally more of the inoculum COD derives from biomass whereas the substrate COD derives from non-degraded substrate. Therefore, it is reasonable to attribute differences in the observed data to differences in the inocula.

3.2. Granular biomass out-performs suspended biomass as inoculum in the BMP assay

For manure, the use of brewery waste AD-granules as the inoculum resulted in a 35% higher biodegradable fraction ($p < 0.05$) than manure AD-sludge (Fig. 1c). The use of paper waste AD-granules also resulted in a higher biodegradable fraction than with manure AD-sludge; however, the difference was not significant (Fig. 1c). Thus, it seems apparent that the source of granules (i.e., brewery waste granular sludge vs. paper waste granular sludge) as an inoculum plays a significant role in determining the biodegradable fraction of manure substrate. It is puzzling, however, that the paper waste AD-granules performed worse than the brewery waste AD-granules, the former being more acclimated to degrade lignocellulosic components, which are the primary constituents of manure (Labatut et al., 2011). The brewery waste AD-granules not only performed better in terms of biodegradability than the paper waste AD-granules for manure substrate, but also yielded a higher maximum methane production rate of $6.0 \text{ mL g COD}^{-1} \text{ d}^{-1}$, which was 40% higher than paper waste AD-granules ($4.3 \text{ mL g COD}^{-1} \text{ d}^{-1}$) (Table S2). On the other hand, both t_{50} and t_c were longer for brewery waste AD-granules (10.6 and 4.8 d, respectively) compared to paper waste AD-granules (4.5 and 0.7 d, respectively, coefficient table is given in Supplementary data). That the granular biomass performed better than the suspended biomass is not surprising, considering that the physical characteristics of the granules can protect the internal microbial community, including methanogens, from the various inhibitory intermediates of AD and also from the fluctuations in pH (Song et al., 2010). In addition, the manure inoculum brings in additional ammonia that increases the already high concentrations of this intermediary in manure substrate.

For sludge, the improvement in biodegradability with both granular biomass source as inocula was considerable and consistent compared to the suspended biomass inocula (Fig. 1d). From a biomethane potential perspective, when sludge was inoculated with granular biomass, both granular biomass inocula showed an approximate 50% higher biomethane yield ($p < 0.05$) than waste activated AD-sludge, without a significant difference between brewery waste and paper waste AD-granules (Fig. 1d). Maximum production rates for both granules were similar ($\mu = 3.8\text{--}4 \text{ mL g COD}^{-1} \text{ d}^{-1}$), which was 19–25% faster than the rate of waste activated AD-sludge. However, both t_{50} and t_c were longer for brewery waste AD-granules (7.0 and 3.0 d) compared to the paper waste AD-granules (4.3 and 0.2 d) as well as the waste activated AD-sludge (5.6 and 2.3 d) (coefficient table is given in Supplementary data).

The advantage of using anaerobic granular biomass, in a BMP assay to inoculate substrates, even those with high solids content such as dairy manure and waste activated sludge, is likely related to the protection of acetogens and methanogens within the granular matrix (Zheng et al., 2006), and also the higher density of these slow-growing microbes compared to any suspended inoculum (Song et al., 2010). Furthermore, the hydrogen transfer rates between species is likely enhanced in anaerobic granules, which therefore may improve biomethane formation (Díaz et al., 2006; Schmidt and Ahiring, 2000). We suspect that the products of hydrolysis can be consumed faster by granules, especially in a broth with breakdown products that may be inhibiting.

3.3. Co-inoculation with suspended AD-sludge and granular biomass does not improve biomethane potential

From the results of the BMP assays, it is clear that brewery waste AD-granules achieved the highest biodegradable fraction for both substrates (Fig. 1). For manure, inoculation with brewery waste AD-granules also resulted in the fastest production rate (coefficient table is given in Supplementary data). Generally, granular sludge tends to have higher biodegradation rates when compared to suspended AD-sludge (Díaz et al., 2006; Schmidt and Ahiring, 2000). It is not clear, however, which factors contributed to the observed differences. To determine whether

co-inoculation of both suspended and granular sludge may perform better due to an improved hydrolysis rate, each substrate was co-inoculated with its respective suspended AD-sludge and AD-granules.

For manure, co-inoculation with both AD-granules resulted in a similar biodegradable fraction compared to manure AD-sludge (Fig. 1e). In fact, only a 7% higher yield was obtained when manure was co-inoculated with brewery waste AD-granules, and a 9% lower yield when it was co-inoculated with paper waste AD-granules. In terms of kinetics (coefficient table is given in Supplementary data), co-inoculation did not perform better than the use of suspended AD-sludge alone. Production rates for both co-inoculation scenarios were $4.7\text{--}5.0 \text{ mL g COD}^{-1} \text{ d}^{-1}$, which was similar to the rate of inoculation with manure AD-sludge. The lag-times were longer for co-inoculation with brewery waste AD-granules and just slightly shorter for co-inoculation with paper waste AD-granules compared with the manure AD-sludge.

For sludge, co-inoculation of suspended AD-sludge with brewery waste AD-granules did show a higher biodegradable fraction, which was 25% higher than inoculation with waste activated AD-sludge (Fig. 1f). Furthermore, adding brewery waste AD-granules to waste activated AD-sludge increased the production rates by 38% (coefficient table is given in Supplementary data). Yet, both t_{50} and t_c for co-inoculation were longer compared to waste activated AD-sludge inoculation (3.2 and 7.5 d, compared to 2.3 and 5.6 d, respectively). The use of paper waste AD-granules biomass to co-inoculate the waste activated AD-sludge performed poorly, resulting in only a 5% higher biodegradable fraction (Fig. 1f). The production rate did not improve by co-inoculation of waste activated AD-sludge with paper waste AD-granules, and the lag-time values were similar. In general, we could not find a clear advantage to co-inoculate the suspended sludge with granular biomass. In fact, we found that inoculating with only granules improved the biomethane potentials the most, and co-inoculation lowers performance by effectively diluting the concentration of granular biomass added. For sludge, inoculation with brewery waste AD-granules provided the highest biomethane potential ($236.2 \text{ mL CH}_4 \text{ g COD}^{-1}$) and paper waste AD-granules provided the shortest lag-phase ($t_c = 0.2 \text{ d}$). All co-inoculation scenarios provided poorer performances that were not necessarily better than suspended inoculation.

It should be noted that the BMP assay is primarily intended to characterize final substrate biodegradability and is not the ideal assay to measure biodegradation rates. However, because each BMP assay in this study was operated under the same initial substrate COD concentration and inoculum to substrate ratio, the rates can be compared to some extent. Our results, well-fitted to a sigmoidal regression model, provide an important insight into the degradation rate and corresponding lag-time for each inoculum, and it is worth noting the overall superior performance of granular biomass.

4. Conclusions

This study emphasized the role of granular biomass as a preferable inoculum in a BMP assay. This conclusion is limited for the tested scenarios since it is unfeasible to evaluate all possible scenarios. Still, differences between inocula of the same type were observed based on inoculum origin, thereby, calling for additional trials using other substrate-inoculum combinations to determine the transferability of our results. Our modification of the Gompertz model provides, in addition to the regular biochemical parameters, an additional parameter (t_c) that allows the comparison of specific aspects of BMP that are critical to characterize the methanization process.

CRedit authorship contribution statement

Roy Posmanik: Investigation, Writing - review & editing. **Andrew H. Kim:** Investigation, Writing - review & editing. **Rodrigo A. Labatut:** Writing - review & editing. **Joseph G. Usack:** Writing - review & editing. **Largus T. Angenent:** Investigation, Writing - review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.biortech.2020.123359>.

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